

Margaret Somerford

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Education:

M.S.M.E. Villanova University: Radnor, PA

August 2013

GPA: 3.67/4.0

Thermo Fluids Certificate

B.S.M.E. Baylor University: Waco, TX

May 2011

GPA: 3.44/4.0

Minor Mathematics

Experience:

Villanova University

2011-Present

Research Assistant: Laboratory for Advanced Thermal and Fluid Systems

- Designed/Analyzed parametric study to optimize the effectiveness of small scale heat exchangers
- Applied ASTM C177-10 to an ICMU. Modeled steady state and transient modes with Comsol
- Graded papers for undergraduate Heat Transfer
- Taught Undergraduate Labs: Milling machine & Surface Roughness

Baylor University– Undergraduate Research Assistant and Class Projects

2009-2011

- Flow visualization and Surface Shear Measurements Downstream of Obliquely Aligned Element Arrays
- Baylor SAE BAJA: Proposed/Lead the Baja Senior Design Project – Front Suspension Design
- Designed and constructed a small scale Recycling Machine: 93% successful

Study Abroad – Baylor i5 Immersion Into International Interdisciplinary Innovation

Summer 2010

- Experience Gain: International Business, Multi-Cultural Teamwork, Analysis of Foreign Market, Business Plan, Presentation

Celebrating Home Distribution Center, Marshall, Texas: Engineering Internship

Summer 2009

- Rack optimization design, Generated ideas to increase picking efficiency, Data Entry

Publications

M.S. Thesis: Thermal Analysis of an Integrated Masonry Unit and a Model for the Effective R-Value

Conference Paper Submissions

- Modeling and Measurements of a Novel Energy Efficient Masonry Wall Unit with Significant Mass and an Integrated Thermal Barrier
- The Effectiveness of Small Scale Heat Exchangers for Electronics Thermal Management

Relevant Skills:

Software: Labview: Created VIs for use with Agilent and NI DAQ systems

Comsol: Steady state and transient heat transfer simulations

SolidWorks and SolidWorks CFD Simulator, MATHCAD, MATLAB, Microsoft Office

Courses: Thermal Management of Electronics, Advanced Engineering Analysis, Advanced Engineering Thermodynamics, Renewable Energy Systems, Advanced Fluid Mechanics, Conduction Heat Transfer, Multiphase Flow and Heat Transfer, Convection

Activities/Service:

- SAE International – President and Leader of initial SAE Baja Project at Baylor
Grew Memberships by 200%; Worked with a team to design the front suspension;
Formed relationship with Letourneau University's Baja Team and partnered with them in our transition/start-up phase.
- A.S.M.E Member, Pi Tau Sigma (Mechanical Engineering Academic Fraternity), Pi Mu Epsilon (Mathematics Honor Society)
- Baylor Student Foundation
- Stepping Out – Baylor Community Service Projects; Habitat For Humanity